

نظم التشغيل

10:30-11:30

29/4/2022

أ.د/خالد فتحي

Faculty of Computers & Information, Assiut University

Course code: CS321

3rd Level

Midterm Exam

Duration: 1 hour

* Required

* This form will record your name, please fill your name.

1

* الإسم الرباعي (بالعربي فقط)

2

* رقم الجلوس

3

* المستوى

- ☐ الاول
- ☐ الثاني
- ☐ الثالث
- ☐ الرابع 2013
- ☐ الرابع 2014
- ☐ الرابع 2015
- ☐ الرابع 2016
- ☐ الرابع 2017
- ☐ الرابع 2018

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* البرنامج

- ☐ عام
- ☐ بايو
- ☐ هندسة

5

* رقم الصالة

☐ أ_1

☐ أ_2

☐ أ_3

☐ أ_4

☐ أ_5

☐ ج_1

☐ ج_2

☐ ج_3

☐ ج_4

☐ ج_5

☐ ج_6

6

* رقم الكمبيوتر

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

8

There are five processes A to E to run. Their arrival times are 0, 1, 3, 9, and 12 milliseconds, respectively. Their processing times are 3, 5, 2, 5, and 5 milliseconds, respectively. What is the turnaround time of process C using first-come-first-served scheduling?

(1 Point)

- ☐ 3
- ☐ 6
- ☐ 7 correct
- ☐ 8
- ☐ 9
- ☐ 10
- ☐ 11
- ☐ 20

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Suppose that a process is in "BLOCKED" state waiting for some I/O service. When the device is completed, it goes to the

(1 Point)

- ☐ BLOCKED state
- ☐ READY state
- ☐ SUSPENDED state
- ☐ TERMINATED state

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A semaphore used as a mutual exclusion lock by a pair of processes should be initialized to

(1 Point)

- ☐ 0
- ☒ 1 correct
- ☐ 2
- ☐ Others

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The following system of four processes with two resources:

Current allocation matrix:

P1 1 3

P2 4 1

P3 1 2

P4 2 0

Current request matrix:

P1 1 2

P2 4 3

P3 1 7

P4 5 1

If the availability vector is [2 4], is the system above deadlocked?

(1 Point)

- ☐ Yes
- ☒ No

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The list of processes waiting for a particular I/O device is called a(n) ____.
(1 Point)

- ☐ standby queue
- ☒ device queue
- ☐ ready queue
- ☐ interrupt queue

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A situation where several processes access and manipulate the same data concurrently and the outcome of the execution depends on the order in which the access takes place, is called a
(1 Point)

- ☐ convoy effect
- ☐ disorder problem
- ☐ critical-section problem
- ☒ race condition

The following system of four processes with two resources:

Current allocation matrix:

P1 1 3

P2 4 1

P3 1 2

P4 2 0

Current request matrix:

P1 1 2

P2 4 3

P3 1 7

P4 5 1

If the availability vector is [2 3], is the system above deadlocked?

(1 Point)

☐ Yes

☐ No

The _____ refers to the number of processes in memory

(1 Point)

☐ process count

☐ long-term scheduler

☐ degree of multi-programming

☐ CPU scheduler

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Consider the following pseudo-code:

```
boolean flag[2];
flag[0]=flag[1]=false;
proc(int i) {
    while(true) {
        while(flag[(i+1) mod 2] == true);
        flag[i]= true;
        critical_section;
        flag[i] = false;
    }
}
```

does the previous pseudo-code solve the critical section problem for two processes?

(1 Point)

☐ Yes

☒ No

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Bootstrap program is kept in the dynamic RAM memory of the computer

(1 Point)

☐ True

☒ False

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test_and_set is applicable to any number of processes not just two processes

(1 Point)

☒ True

☐ False

There are five processes A to E to run. Their arrival times are 0, 1, 3, 9, and 12 milliseconds, respectively. Their processing times are 3, 5, 2, 5, and 5 milliseconds, respectively. What is the turnaround time of process E using Shortest-job-first scheduling?

(1 Point)

- ☐ 3
- ☐ 6
- ☐ 7
- ☒ 8 correct
- ☐ 9
- ☐ 10
- ☐ 11
- ☐ 20

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```
lock = 0;  
R=compare_and_swap(&lock, 0,1);
```

The values of lock and R after executing the previous pseudo-code are:
(1 Point)

- ☐ lock =0, R=0
- ☐ lock =0, R=1
- ☒ lock =1, R=0
- ☐ lock =1, R=1
- ☐ Others

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There are five processes A to E to run. Their arrival times are 0, 1, 3, 9, and 12 milliseconds, respectively. Their processing times are 3, 5, 2, 5, and 5 milliseconds, respectively. What is the turnaround time of process B using round-robin (with 1 second quantum) scheduling?

(1 Point)

- ☐ 3
- ☐ 6
- ☐ 7
- ☐ 8
- ☐ 9
- ☐ 10 correct
- ☐ 11
- ☐ 20

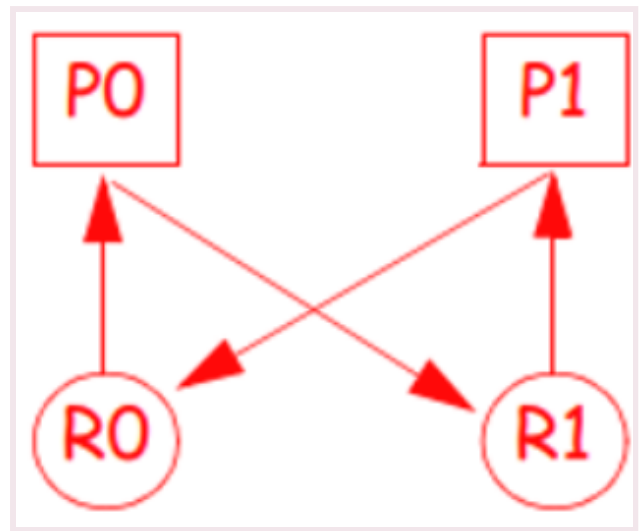
22

Critical section is

(1 Point)

- ☐ a part of the operating system which is not allowed to be accessed by any process
- ☐ a set of instructions that access common shared resource which exclude one another in time
- ☐ a part of the kernel that interfaces to the device controllers
- ☐ none of the above

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The two processes in following resource allocation graph are:

(1 Point)

- ☐ **deadlocked**
- ☐ Not deadlocked

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In the readers/writers problem we want to allow multiple writers to be in their critical sections for the shared data at the same time.

(1 Point)

- ☐ True
- ☐ **False**

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The two separate modes of operating in a system are
(1 Point)

- ☐ supervisor mode and system mode
- ☐ kernel mode and privileged mode
- ☐ physical mode and logical mode
- ☐ user mode and kernel mode

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The two fundamental models of interprocess communication are message passing and shared memory
(1 Point)

- ☐ True
- ☐ False

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The cache is managed by
(1 Point)

- ☐ Compiler
- ☐ Hardware
- ☐ Operating system
- ☐ Others

